# Evaluation report

*Generated 2026-10-12 14:33 UTC. Out-of-sample results from purged walk-forward validation (docs/design.md section 7).*

## Key findings

- Out of sample: 86 release slots from 2026-05-28 to 2026-10-08, each traded at the first price after the release, with 2 bps of costs per side.
- Best Sharpe ratio: Ridge, 1.27 (90% interval -1.39 to 4.51), Sortino 1.87, max drawdown -7.9%, total return 6.6%.
- Other Sharpe ratios: Naive baseline (strategy mean) 1.16; Gradient boosting -1.14; Always long SPY, 1 session -0.30; SPY buy and hold 0.84.
- Strategy ranking, rank IC: baseline 0.124 (t = 3.38); ridge 0.021 (t = 0.40); hgb -0.028 (t = -0.56). Only |t| of 2 or more marks a reliable ranking.
- Win-rate forecasts: AUC 0.49 to 0.55 (0.50 is a coin flip).
- Every 90% Sharpe interval includes zero, so with this sample no strategy's edge is distinguishable from noise.

## Data coverage

### Events by category

| category | events | with_actual | first | last |
|---|---|---|---|---|
| consumer | 21 | 18 | 2026-04-24 | 2026-11-06 |
| fed | 33 | 4 | 2026-04-17 | 2026-10-28 |
| growth | 51 | 43 | 2026-04-15 | 2026-11-04 |
| housing | 4 | 4 | 2026-04-21 | 2026-05-28 |
| inflation | 71 | 57 | 2026-04-30 | 2026-11-10 |
| labor | 69 | 59 | 2026-04-16 | 2026-11-06 |
| other | 34 | 0 | 2026-04-15 | 2026-10-07 |

### Release-time anchors

| event | releases | times_et |
|---|---|---|
| ADP Non-Farm Employment Change | 7 | 08:15 |
| CPI m/m | 7 | 08:30 |
| CPI y/y | 7 | 08:30 |
| Core CPI m/m | 7 | 08:30 |
| Core CPI y/y | 7 | 08:30 |
| Federal Funds Rate | 5 | 14:00 |
| Non-Farm Employment Change | 7 | 08:30 |

### News by provider and theme

| provider | theme | articles | first | last |
|---|---|---|---|---|
| exa | fed | 244 | 2026-04-15 | 2026-10-12 |
| exa | growth | 242 | 2026-04-15 | 2026-10-11 |
| exa | inflation | 249 | 2026-04-15 | 2026-10-12 |
| exa | labor | 256 | 2026-04-16 | 2026-10-10 |
| tavily | fed | 9 | 2026-10-06 | 2026-10-11 |
| tavily | growth | 3 | 2026-10-05 | 2026-10-11 |
| tavily | inflation | 6 | 2026-10-06 | 2026-10-10 |
| tavily | labor | 5 | 2026-10-08 | 2026-10-09 |

### Prices by symbol

| symbol | sessions | first | last |
|---|---|---|---|
| GLD | 164 | 2026-02-17 | 2026-10-09 |
| SPY | 164 | 2026-02-17 | 2026-10-09 |
| TLT | 164 | 2026-02-17 | 2026-10-09 |
| UUP | 164 | 2026-02-17 | 2026-10-09 |
| ^VIX | 164 | 2026-02-17 | 2026-10-09 |

### Slots and trades

| slots | slots_with_surprise | panel_rows | oos_slots |
|---|---|---|---|
| 117 | 85 | 4736 | 86 |

## Walk-forward set-up

- **As-of date:** 2026-10-12
- **Window:** 2026-04-15 to 2026-11-11
- **Sessions in the calendar:** 164
- **Minimum training slots:** 30
- **Walk-forward step:** 1
- **Out-of-sample period:** 2026-05-28 to 2026-10-08
- **Folds:** 86
- **Models:** regressors baseline, ridge, hgb; classifiers baseline_clf, logit, hgb_clf
- **Selection:** each slot trades the strategy with the highest predicted net return when it is above zero, otherwise stays flat
- **Costs:** 2.0 bps per side
- **Purging:** a fold trains only on trades that exited before the test slot's entry

## Predictive accuracy

| model | target | n_slots | n_preds | rank_ic | rank_ic_t | hit_rate | mae | auc | brier |
|---|---|---|---|---|---|---|---|---|---|
| baseline | net_ret | 86 | 3,536 | 0.124 | 3.38 | 55.5% | 1.01% | n/a | n/a |
| ridge | net_ret | 86 | 3,536 | 0.021 | 0.40 | 49.2% | 1.04% | n/a | n/a |
| hgb | net_ret | 86 | 3,536 | -0.028 | -0.56 | 48.9% | 1.07% | n/a | n/a |
| baseline_clf | win | 86 | 3,536 | 0.101 | 3.19 | 56.2% | n/a | 0.554 | 0.255 |
| logit | win | 86 | 3,536 | -0.036 | -0.92 | 50.1% | n/a | 0.490 | 0.257 |
| hgb_clf | win | 86 | 3,536 | -0.014 | -0.29 | 50.4% | n/a | 0.505 | 0.263 |

Rank IC: mean per-slot Spearman correlation between predicted and realised net return across strategies. Hit rate: share of predictions with the right sign (classifiers: probability above 0.5 matches the outcome). AUC and Brier score apply to the win classifiers.

## Strategy performance

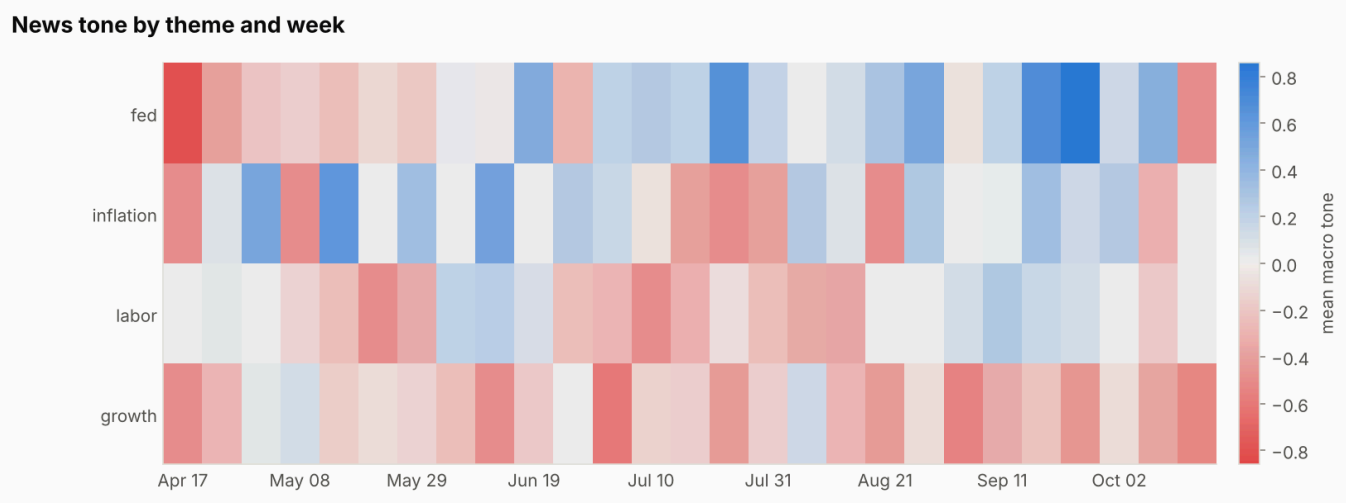
| strategy | sharpe | sharpe_lo | sharpe_hi | sortino | max_drawdown | total_return | cagr | ann_vol | exposure | n_days | n_trades | win_rate | avg_return | profit_factor |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Naive baseline (strategy mean) | 1.16 | -1.13 | 3.87 | 1.91 | -9.8% | 8.6% | 25.1% | 21.3% | 100.0% | 93 | 86 | 55.8% | 0.38% | 1.33 |
| Ridge | 1.27 | -1.39 | 4.51 | 1.87 | -7.9% | 6.6% | 18.8% | 14.3% | 86.0% | 93 | 86 | 59.3% | 0.16% | 1.46 |
| Gradient boosting | -1.14 | -4.56 | 1.48 | -1.56 | -8.2% | -5.7% | -14.8% | 13.2% | 92.5% | 93 | 86 | 46.5% | -0.30% | 0.67 |
| Always long SPY, 1 session | -0.30 | -2.99 | 1.67 | -0.43 | -4.2% | -1.2% | -3.3% | 9.7% | 66.7% | 93 | 86 | 44.2% | -0.05% | 0.84 |
| SPY buy and hold | 0.84 | -1.76 | 3.07 | 1.25 | -4.5% | 3.6% | 10.2% | 12.5% | 100.0% | 93 | n/a | n/a | n/a | n/a |

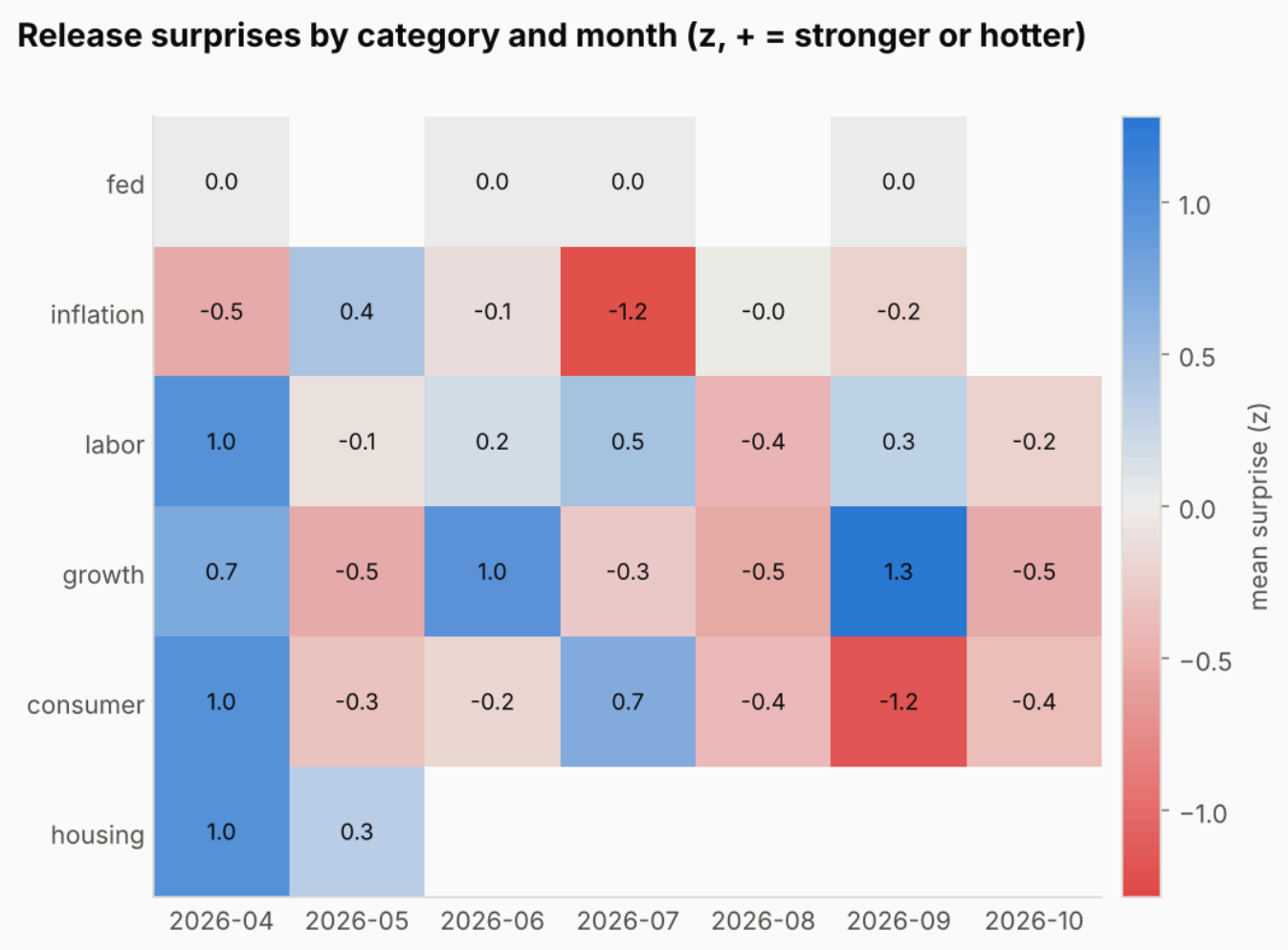
Sharpe interval: 90% moving-block bootstrap (blocks of 5 sessions). Metrics use daily calendar-time returns, 252 sessions a year, risk-free rate 0.

## Best strategies out of sample

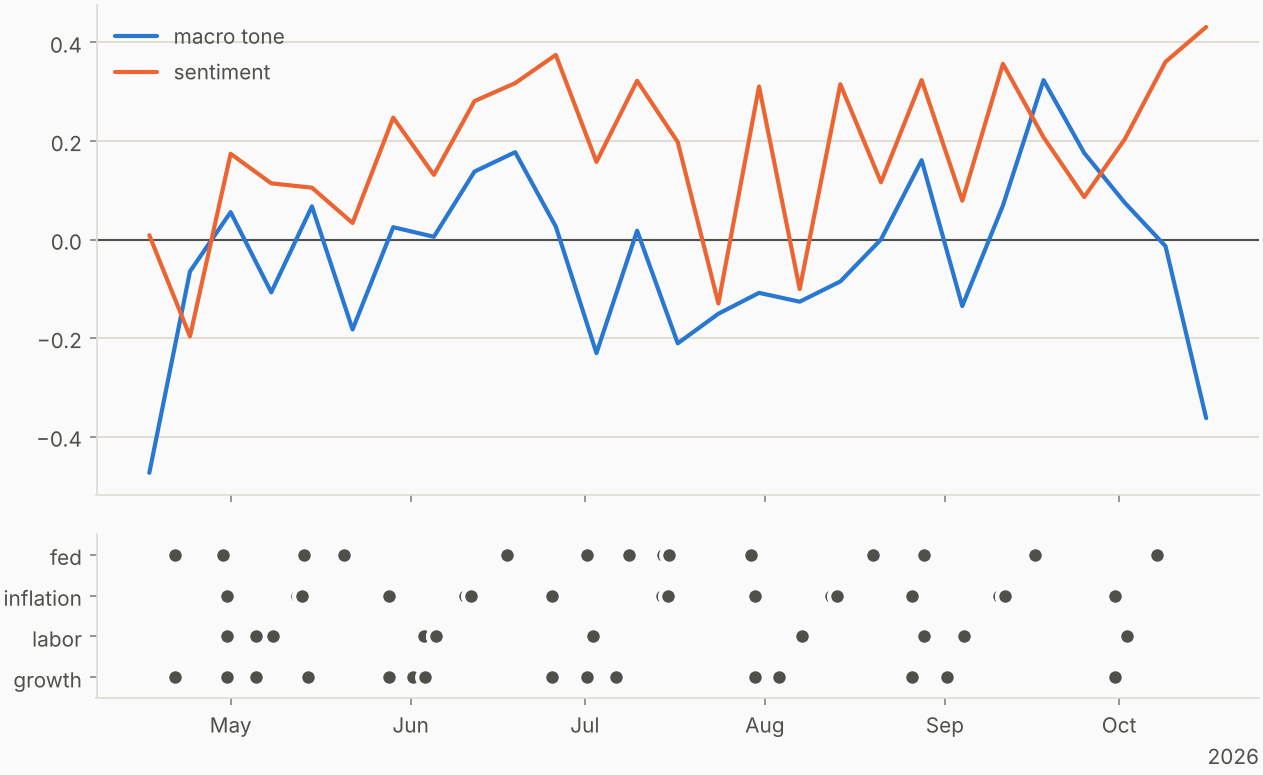
| strategy_id | mean_ret | win_rate | n |
|---|---|---|---|
| GLD\|short_on_beat\|5 | 0.72% | 64.5% | 62 |
| GLD\|short\|5 | 0.49% | 56.6% | 83 |
| GLD\|short_on_beat\|3 | 0.46% | 54.0% | 63 |
| TLT\|short\|5 | 0.45% | 66.3% | 83 |
| TLT\|short\|3 | 0.27% | 61.9% | 84 |
| UUP\|long\|5 | 0.24% | 68.7% | 83 |
| GLD\|short_on_beat\|1 | 0.22% | 50.0% | 64 |
| SPY\|short_on_beat\|3 | 0.22% | 57.1% | 63 |
| GLD\|short\|3 | 0.20% | 60.7% | 84 |
| UUP\|long_on_beat\|5 | 0.17% | 58.1% | 62 |

## Figures

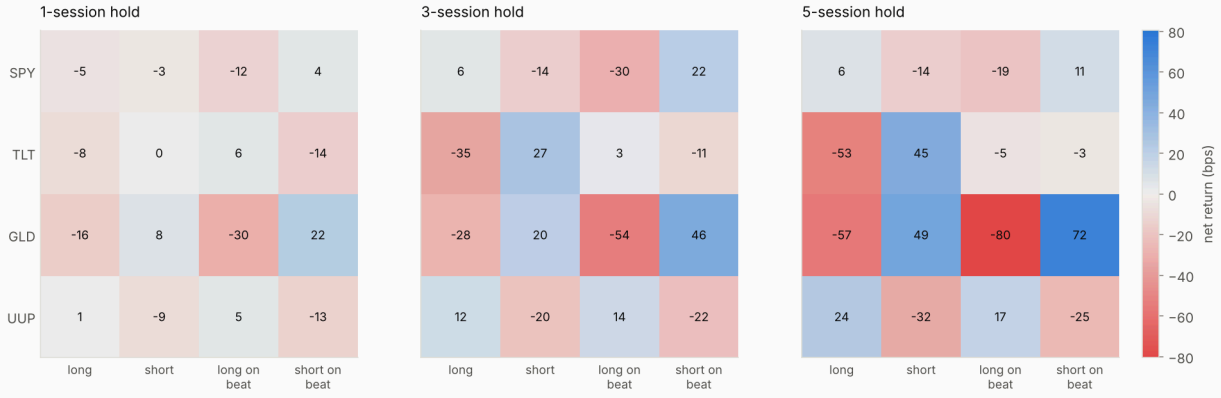

News tone by theme and week
fed
inflation
labor
growth
Apr 17
May 08
May 29
Jun 19
Jul 10
Jul 31
Aug 21
Sep 11
Oct 02
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
mean macro tone


Release surprises by category and month (z, + = stronger or hotter)
fed
inflation
labor
growth
consumer
housing
0.0
0.0
0.0
0.0
-0.5
0.4
-0.1
-1.2
-0.0
-0.2
1.0
-0.1
0.2
0.5
-0.4
0.3
-0.2
0.7
-0.5
1.0
-0.3
-0.5
1.3
-0.5
1.0
-0.3
-0.2
0.7
-0.4
-1.2
-0.4
1.0
0.3
2026-04
2026-05
2026-06
2026-07
2026-08
2026-09
2026-10
1.0
0.5
0.0
−0.5
−1.0
mean surprise (z)

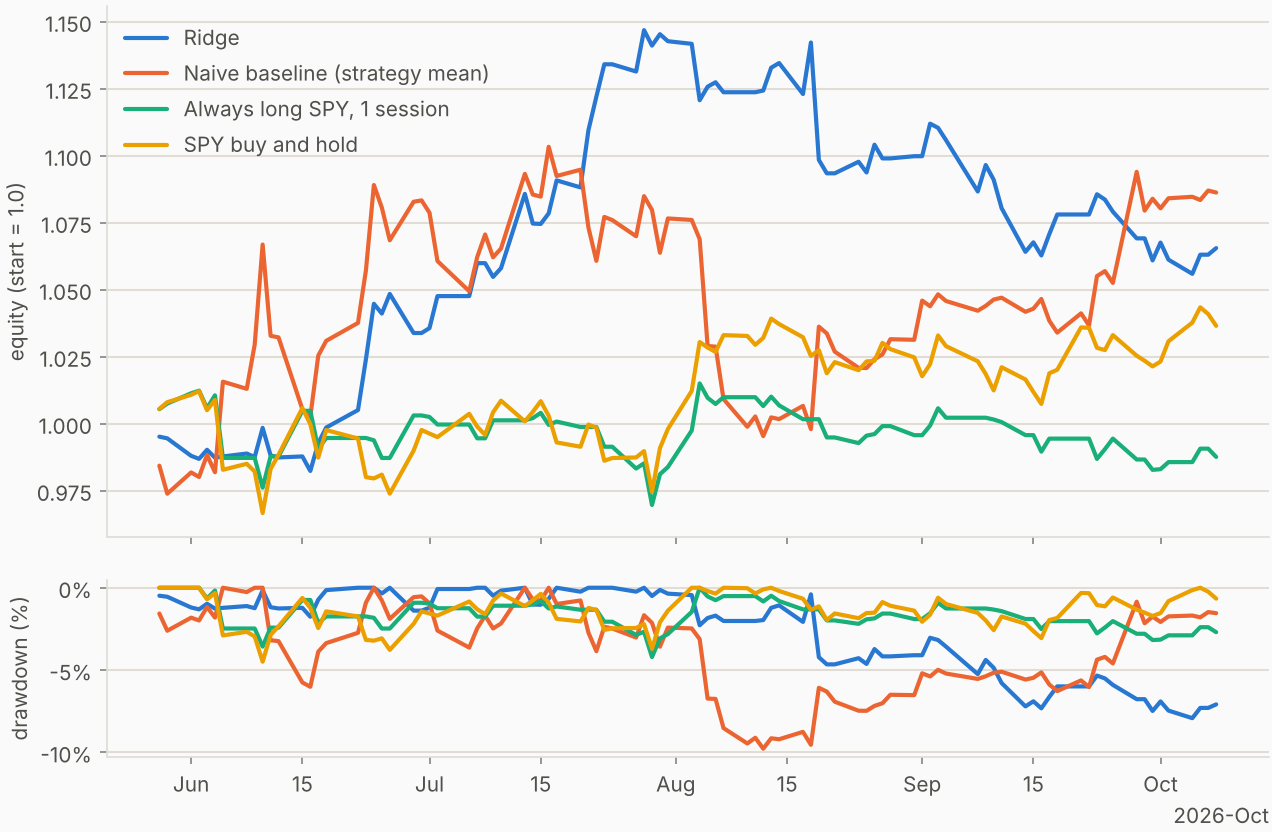
## Macro news sentiment and high-impact releases




**Out-of-sample mean net return per trade (bps)**



## Equity and drawdown, out of sample




## Notes

- Articles published after the as-of time: 3
- Events in the window without an actual value (upcoming or speeches): 98

## Limitations

- Small sample: about 180 days gives roughly 100 slots, so out-of-sample results carry wide confidence intervals.
- Daily bars: entry waits for the next open or close, so the immediate reaction to a release is only partly captured.
- Forecasts are the calendar site's consensus, not a survey of record.
- Sentiment from the macro tone lexicon and VADER is coarse.
- Costs are a flat 2 bps per side; slippage around releases can be larger.
- Early market closes are not modelled; a release after a 13:00 ET early close is treated as in-session.